

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 2nd Term Examination, 2022

URP-2281
(GIS and Remote Sensing)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) "The evolution of geotechnology community has broadened its membership in members, interests, background and depth of understanding" – Explain with necessary example. (10)
(b) Why GIS as a separate profession is important? Explain with a focus to Bangladesh. (10)
(c) GIS can be applied to combat waterlogging problem in Khulna City. Discuss the applicability of GIS to different aspects (i.e., location, condition, trend, routing, pattern and modelling) to mitigate the impacts of waterlogging in Khulna City. (15)
2. (a) "Geospatial data is consisted of spatial data, aspatial data and meta data". Explain the statement. (15)
(b) Why are spatial references in data sets important for consistent analysis? Explain with rational examples. (10)
(c) Suppose you have to prepare a map of entire Bangladesh. Will you use Universal Transverse Mercator (UTM) as coordinate system? If so, describe the benefits. If not, discuss the possible alternative. (10)
3. (a) What are the major drawbacks of spatial database infrastructure in Bangladesh? (05)
(b) Suppose you have a paper mouza map of Khulna City. You need to link proposed plans to the mouza map for implementing the priority projects. Is it necessary to georeference the paper map before performing any analysis? If so, describe the process. If not, explain why. (15)
(c) Suppose you have converted the geographic data of your project, either from a hardcopy or a scanned image, into vector data by digitizing the features. What kinds of errors could occur during processing? Discuss in brief with necessary examples and figures. (15)
4. (a) Assume you have physical data (vector data) for Khulna City that you want to use to solve some spatial planning problems. Describe the applicability of different spatial queries/operations for solving the problems in Khulna City. Use practical examples and figures where necessary. (20)
(b) Write short note on the following: (15)
 (i) Topology
 (ii) Shapefile
 (iii) Database Management System

Section – B

5. Dhaka, the capital of Bangladesh, is experiencing rapid urban expansion. Urban planners are using satellite image analysis techniques to detect and manage this growth.
Read the aforementioned statement and answer the following questions.
- (a) Explain the principle that allows remote sensing to distinguish different objects based on their emitted or reflected radiant energy? (10)
 - (b) What else is it possible to measure with remote sensing for Dhaka, besides managing urban expansion? (10)
 - (c) How can satellite image classification techniques help in identifying areas of urban expansion and green space reduction? (15)

6. (a) When do you use True Color Combination (TCC) and False Color Combination (FCC) in remote sensing analysis and why? (10)
- (b) What role does atmospheric correction play in enhancing the accuracy of satellite imagery? (10)
- (c) Describe different types of digital images used in remote sensing. (15)

7. (a) Compare between two sets of satellite Images having different types of resolutions in the following. Explain with proper example. (10)

Satellite Image (A)	Satellite Image (B)
(i) High spatial resolution (0.3 - 4 m)	(i) High spatial resolution (0.3 - 4 m)
(ii) Low spectral resolution (1 - 3 bands)	(ii) High spectral resolution (16 - 220 bands)
(iii) High radiometric resolution (16 bit)	(iii) Medium radiometric resolution (8 bit)

- (b) Describe the concepts of FOV, AFOV, IFOV, and GIFOV in remote sensing. (15)
- (c) Determine the IFOV when height of satellite 904 km and spatial resolution 36 m. (10)

8. Utilizing drone surveys for master plan preparation offers a cutting-edge approach to urban planning. Drones provide high-resolution aerial imagery and topographic data, enabling detailed analysis of land use, infrastructure, and environmental features.

Read the aforementioned statement and answer the following questions.

- (a) Write down the use of orthophoto and stereo model in master plan preparation. (10)
- (b) Assume that the distance between two points was measured to be 83.33 mm on a vertical photograph and 125 mm on a map. If the surveying ground distance between the same two points is 3000 m, what will be the photo scale and map scale and the scale reciprocals of the photograph and the map? (10)
- (c) What challenges might urban planners face when using drone-based photogrammetry for master plan preparation, and how can these be mitigated? (15)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 2nd Term Regular Examination, 2022
URP 2251
(Landscape Planning and Design)

Full Marks: 210

Time: 3.00 hrs.

- N.B.** i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

- 1 (a) Define landscape. (05)
(b) Discuss the importance of studying landscape planning and design. (15)
(c) Distinguish landscape features, landscape elements and landscape principles with appropriate examples. (15)
- 2 (a) What is color scheme? Discuss with example different color schemes. (15)
(b) What is unity? Discuss different types of unity with appropriate example. (10)
(C) Explain proportion used in plant and landscape with necessary sketch. (10)
- 3 (a) Write down the significance of studying human ecology. (11)
(b) Analyze Carbon Cycle with necessary drawings. (12)
(c) Write short note on the followings: (12)
i) Tree forms in landscape
ii) Landscaping with native wildflowers and grasses
iii) Repetition in landscaping.
- 4 (a) Discuss the general challenges, goals, plant characteristics and design solutions with and without trees on any two of the followings: (20)
i) Front entry
ii) Under window
iii) Along sidewalks
(b) Evaluate with example the nine landscape principles should be followed in landscape plan and design. (15)

Section – B

- 5 (a) What is Landscape Site? (05)
(b) Describe the 19th Century Historical Reference of Landscape Design with example. (18)
(c) Write Short Note on: (12)
i. Climate and Regions in Landscape Design
ii. The Renaissance and Baroque- An Awakening of Creativity in Landscape Design
- 6 (a) Why water is considered as a significant resource in Landscape Planning? - Explain (05)
(b) How can we get the maximum benefits from water features in a landscape site? - Explain with appropriate sketches. (15)
(c) Write Short Note on: (15)
i. 20th Century Landscape Design Concept
ii. Land and Law
iii. The Nine Principles of Water-wise Landscaping
- 7 (a) What is landscape conservation? (05)
(b) Explain the principles of landscape conservation. (15)
(c) Discuss different steps to develop a residential landscape design with sketches. (15)
- 8 (a) What is Axis? (05)
(b) Discuss the properties of Axis. (12)
(c) Explain the various "Structure Pattern" in city scale landscape planning and design with sketches. (18)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 2nd Term Regular Examination, 2022
Hum 2217
(Public and Local Government Finance)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

- 1 (a) What do you mean by public finance? (05)
(b) Write down the role of public finance in a developing country like Bangladesh. (20)
(c) What are the basic differences between central and local government finance. (10)
- 2 (a) Evaluate if the burden of indirect tax can be shifted to another person or not. (05)
(b) What elements do you think an economy should have in an ideal tax system? (15)
(c) Discuss the different kinds of tax according to tax rate. (15)
- 3 (a) What are the sources from where government can borrow money? (15)
(b) Why public debt management is important in a country like Bangladesh? (05)
(c) How public debt is different from private debt? (15)
- 4 (a) Why public expenditure is increasing over time in developing countries? (15)
(b) "Peacocks and Wiseman Hypothesis on public expenditure is an improved version of Adolf Wagner's Hypothesis". Justify the statement with relevant explanation. (20)

Section – B

- 5 (a) Define deficit financing. (05)
(b) Distinguish between deficit financing and deficit budgeting. (10)
(c) Mention the effects of deficit financing in a developing country like Bangladesh. (20)
- 6 (a) Write down your understanding about public budget. (05)
(b) Discuss the steps of formulating public budget in a developing country like Bangladesh. (20)
(c) Distinguish between revenue budget and surplus budget. (10)
- 7 (a) What is meant by local government? (05)
(b) Mention the types of local government in Bangladesh. (10)
(c) What is the importance of local government of a country while there is a central government? (20)

- 8 (a) What is city corporation? Mention the structure of city corporation. (10)
- (b) Describe the functions of city corporation as a local government. (15)
- (c) What is the formation system of Zilla Parishad in Bangladesh? (10)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning

BURP 2nd Year 2nd Term Regular Examination, 2022

CE 2241

(Principles of Environmental Engineering)

Full Marks: 210

Time: 3.00 hrs.

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable value for any missing data.

Section – A

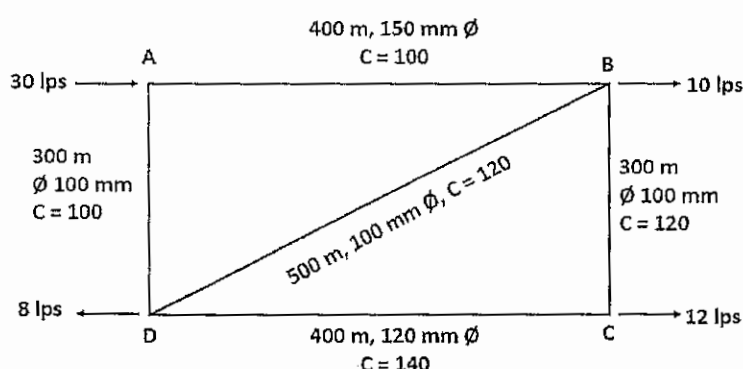
- 1 (a) What is Environmental Engineering? Why planners need to give special attention in environment? (08)
- (b) What are the objectives of water supply? Mention the elements of water supply system and draw the surface water-based water supply system with neat sketch. (10)
- (c) A settling tank has an internal area of 125 m². The particles in the tank have a diameter of 10⁻⁵ m, Re = 0.4, and specific gravity = 2.60. The water temperature in the tank is 298K and the sedimentation period is 218 min. The sludge accumulation zone is 0.5 m and the total depth (including sludge accumulation zone) is 2.1 m. Determine the tank efficiency. If the tank efficiency is less than 90%, redesign the tank to maintain an efficiency of 90%. (17)
- 2 (a) What is LPCD? What are the factors affecting per capita water consumption? (10)
- (b) Mention the characteristics on which drinking water quality depends. Discuss all. (12)
- (c) "The most important parameter of drinking water is the bacteriological quality" - Explain it. (13)
- 3 (a) Define: i. Flocculent Settling ii. Coagulant iii. Break Point Chlorination (10)
- iv. Permanent Hardness v. Disinfection
- (b) The following water samples contain various impurities as shown below. Draw the safest yet cheapest treatment process diagram for each sample and briefly discuss the reason. (12)

Sample-1	Sample-2
Source: Lake; TC: 100 N/100 mL; Color: 20 TCU; Turbidity: 60 NTU; Contain: 50% settleable aggregates and 50% colloidal fine particles with specific gravity < 1.	Source: River; Type: Palatable TC: 1 N/100 mL; Turbidity: 4 NTU Contain: 100% settleable aggregates with average size < 0.5 cm.

- (c) Estimate the population at 2100 and 2120 based on the given data. Use least square parabolic method. (13)

Year	2010	2030	2050	2070	2090
Population (thousand)	300	317	334	380	405

- 4 (a) What are the key components of water safety plan (WSP) introduced by WHO. (07)
- (b) Calculate the flow in each of the pipes of the following looped pipe network. [use at least two trials] (28)



Section – B

- 5 (a) Define Environmental Sanitation. Draw a schematic diagram of disease transmission from excreta and indicate different preventive measures including sanitation. (10)
- (b) Lists some common infectious diseases that are transmitted due to lack of proper sanitation. Write down the transmission pattern of following diseases: (10)
- i. Diarrhoea ii. Ascaris iii. Schistosomiasis.
- (c) The catchment area is of 250 hectares. The surface cover in the catchment is given in below table. Calculate the quantity of storm water runoff, if the rainfall intensity is 35 mm/hr. Also determine the diameter of sanitary sewer as separate system by using Manning's equation for the following information: (15)
- i. Population density in the area 350 person per hectare ii. Rate of water supply is 200 LPCD. iii. Sewer is to be laid on slope 1 in 200 (Assume reasonable missing data.)

Cover Type	Runoff co-efficient	Percentage
Pavement and Yards	0.70	20
Roofs	0.80	10
Lawns and Gardens	0.10	25
Roads	0.40	20
Open Ground	0.15	15
Single Family Dwelling	0.50	10

- 6 (a) What is sewerage system? Distinguish between separate and combined sewerage system. What is self-cleaning and no-scouring velocity? (10)
- (b) What is sanitation? Briefly discuss the sanitation system utilized in Bangladesh. (10)
- (c) A septic tank having dimension 1m x 3m x 1.5m is designed to serve a household of six persons, who produce 75 LPCD of waste water. Design a soakage pit for the disposal of effluent from septic tank. Consider the soil is porous silty clay loam. (15)
- 7 (a) Describe the bacterial growth in waste water with neat sketch. Briefly describe the symbiosis between algae and bacteria in facultative pond. (10)
- (b) Design an anaerobic, facultative and maturation pond system to treat waste water from a community with 80,000 population and flow 150 LPCD. The incoming waste water has the $BOD_5 = 600,000 \text{ mg/m}^3$; temperature = 20°C , $N_i = 3.5 \times 10^7 / 100 \text{ mL}$. It is desired that treated effluent is used in irrigation at 25°C and permissible faecal coliform is less than 6000/100 mL. (15)
- (c) Describe different methods of solid waste collection system. In case of Khulna City, which method is more appropriate and why? (10)
- 8 (a) Explain the following terms with example: (08)
 - i. Reduce
 - ii. Recycle
 - iii. Reuse
 - iv. Recover
- (b) Write short notes on-
 - i. Ocean Acidification
 - ii. Noise Barrier
 - iii. Air Quality Index
 - iv. Surface Runoff
 - v. Clean Air(10)
- (c) Give some examples of noise levels with their sources and subjective evaluation. Describe the effects and control measures of soil pollution. (10)
- (d) Calculate the volume of landfill area needed to serve a population of 20,000 for a period of 20 years. The solid waste generation rate is 2 kg/person/day. The density of uncompacted solid waste is 100 kg/m^3 and compaction ratio of 3.5 is suggested. The ratio of compacted fill (SW + cover) to compacted solid waste is 1.5. (07)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 2nd Term Examination, 2022
URP 2291
(Statistics For Planners - II)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Necessary charts and graphs will be provided (F distribution, T distribution, Z table, Chi-square table).
iv) Assume data where necessary.

Section – A

1. (a) What is the difference between parametric test and non-parametric tests? Explain in short. (08)
- (b) "Macro-data in nature has a tendency to be normally distributed" – explain with reference to 68-95-99 rules. (09)
- (c) Write short notes on the following: (18)
 - i. Null Hypothesis Vs Alternative Hypothesis
 - ii. One tailed Vs Two tailed tests.
 - iii. Critical significance Vs Practical significance
2. (a) Graphically explain T-distribution, Z-distribution and F-distribution. (08)
- (b) The United Nations Development Program (UNDP) thinks that their project on slum dwellers in Khulna city has positive income effect between 2010 and 2023. The UNDP collected two sets of random samples from normally distributed incomes. (12)

Sl no.	Weekly income (in \$ US), 2010	Weekly income (in \$ US), 2023
1	12	15
2	25	29
3	23	32
4	10	18
5	17	28
6	18	20
7	22	22
8	32	45
9	22	18
10	9	

Do you think that the UNDP's assumption is true? Explain with appropriate test.

- (c) Mean lifetime of a sample of 100 light bulbs is computed to be 1750 hours. If the mean lifetime of all the bulbs produced by a company is known to be 1600 and standards deviation to be 1200 hours. Test the hypothesis that samples came from this company. (15)
3. (a) What is Test of the Goodness of Fit? Explain with example. (10)
- (b) Five coins were tossed 100 times and the number of time tails were shown are given below: (15)

Number of tails (T)	0	1	2	3	4	5
Frequency of toss show tails (T)	3	16	32	30	15	4

Can you say that the coins are unbiased?

- (c) Out of 800 workers in a factory 300 are literates. 100 workers were screened out (10)
due to inefficiency out of which 20 were literates. Test as to whether the efficiency
of the worker is associated with education.

Two-way frequency table showing observed frequency

	Efficient	Inefficient	Total
Literate	280	20	300
Illiterate	420	80	500
Total	700	100	800

4. (a) Following table shows the presence of Cancer cells at initial stage and after two (20)
weeks of 15 sample patients. If the samples are chosen from a normally
distributed population, is there a significant difference between the two-stages.

Sl no.	Cancer-Initial Stage	Cancer-After two weeks
1	6	6
2	9	6
3	7	9
4	6	7
5	6	7
6	6	6
7	6	11
8	6	11
9	6	9
10	6	4
11	7	8
12	6	6
13	8	8
14	7	16
15	6	10

- (b) Ten students of KUET are asked to rank (out of 10) three (A, B, C) drinks based (15)
on their tests. Is there a significant difference among the three drinks?

Sl no.	A	B	C
1	4	6	5
2	7	8	7
3	6	5	5
4	9	8	5

Section – B

5. Imagine you are a researcher investigating the relationship between the production skills and
income levels of planning students. You have collected scores for both production skills and
income for a group of planning students, as shown in the table below. Your goal is to explore
whether there is a correlation or not.

Productivity Score	Income Level				
	20000	30000	40000	50000	60000
40-50	3	6	5	2	2
50-60	4	7	4	3	4
60-70	2	2	4	5	3
70-80	2	4	3	0	4
80-90	1	2	2	3	6
90-100	2	0	4	6	5

- (a) Calculate the correlation coefficient between the Productivity Score and Income (15)
Level for the planning students?

- (b) Interpret the correlation coefficient to understand the strength and direction of the relationship between production skills and income. (08)
- (c) Make a comparative analysis of the approaches that can be applied to examine this type of Correlation? (12)
6. (a) A coefficient of rank correlation between score of two judge is found to be 0.413. If the sums of squares of the differences in rank is given to be 64. Find the total number of participants. (11)
- (b) Illustrate the differences between Correlation and Regression with example. (07)
- (c) "Though correlation coefficient is independent of change of scale but regression coefficient is not independent of change of scale" – Draw/prove your argument regarding this statement. (17)
7. (a) Suppose, you are the Transport Planner of Khulna City. You overserved that 6 Zones of the city has the no. of Bus station in relation to the population in 1000 was follows: (18)

No. of Bus station	Population
1	2
6	7
4	3
5	5
2	8
3	4

- i) Find the linear equations?
- ii) Find Y in terms of X?
- iii) Find X in terms of Y?
- iv) Find the value of coefficient of determination?
- (b) What does the value of R^2 denotes – Explain with examples. (07)
- (c) Discuss the concept of Least square method with the help of diagram. (10)
8. (a) Below are given figures of sales (in lakhs) of a Toy factory (18)

Year	Sales
2008	80
2009	92
2010	99
2011	84
2012	88
2013	94
2014	99

- i) Fit a straight-line trend to these figures.
- ii) Plot these figures on a graph and show the trend line.
- iii) Estimates the likely sales of the company during 2016.
- (b) What are the adjustments needs to make before time series analysis? Explain with examples. (07)
- (c) Briefly explain the components of time series with necessary diagram. (10)